PREMIER LEAGUE

VAR Impact Analysis

An Exploratory Data Analysis & Interactive Power BI Dashboard — 2019/20 Season

Data source: FIFA VAR Incidents dataset (Kaggle)
Tools: Power Query (M) · Power BI Desktop
Scope: 20 clubs · 218 logged VAR incidents

Team: -

- **Taha Mohammed Amer (20240370)**
- **Ahmed Hamdy Samir El-Batanony (20240109)**
- **Mostafa Mahmoud EL-Sayed (20249612)**
- **Ali Abdul-Rasoul Salama (20240024)**
- **Mohammed Farhat Mohammed Diab (20240476)**

Course: Business analysis	**DR/ Hatem Abdelkader**	**ENG/ Mohammed Safwat**

1. Data Overview

This report examines Video Assistant Referee (VAR) activity across the 2019/20 Premier League season, using the FIFA VAR Incidents dataset published on Kaggle. The dataset logs every VAR review recorded that season and how it affected the match outcome, organised across two related tables that were connected in Power BI on the Team field.

1.1 Dataset structure

Table	Grain	Rows	Purpose
VAR_Incidents_Stats	One row per VAR review	218	Every individual VAR incident: the team and opponent involved, venue, match date, a free-text description, the match minute, and whether the review ultimately worked for or against the team.
VAR_Team_Stats	One row per club	20	Pre-aggregated, club-level totals — how many reviews a club was involved in, how many led to or disallowed a goal for/against it, the resulting net goal impact, and a narrower breakdown of purely judgment-based ("subjective") decisions.

The two tables are connected one-to-many (VAR_Team_Stats → VAR_Incidents_Stats) on Team, with the cross-filter direction set to Both, so that fields which only exist in the incident-level table — such as Incident Type and Site — can still filter the club-level aggregates on the dashboard.

1.2 Column glossary — VAR_Incidents_Stats

Column	Description
Team / Opponent Team	The two clubs involved in the match.
Date	Match date (originally without a year — see Section 2.1).
Site	H = home fixture for Team, A = away fixture.
Incident	Free-text description of what happened and why VAR intervened.
Incident type	Category derived from Incident during cleaning (Penalty, Red Card, Offside, Handball, Foul, Other).
Minute	The match minute the review occurred, extracted from the original Time text.
VAR used	FOR or AGAINST — whether the review's outcome favoured Team.

1.3 Column glossary — VAR_Team_Stats

This table required no cleaning: every column was already a complete numeric aggregate. The four goal-impact columns are the ones most often misread, so they are defined precisely below:

Column	Meaning
Overturns	Total number of VAR reviews the club was involved in (equal to its row count in VAR_Incidents_Stats).
Leading to goals for	A review resulted in a goal being awarded to the club that would not have counted otherwise.
Disallowed goals for	A review disallowed a goal scored by the opponent — a benefit to the club even though the club didn't touch the ball.
Leading to goals against	A review resulted in a goal being awarded to the opponent that would not have counted otherwise.
Disallowed goals against	A review disallowed a goal the club itself had scored.
Net goal score	The net effect of the four columns above: positive means VAR helped the club overall; negative means it hurt the club overall.
Subjective decisions for / against	The subset of decisions resting on human judgement (e.g. foul, handball, red card) rather than a fact-based check like offside.
Net subjective score	Subjective decisions for minus subjective decisions against.

2. Analysis Process

Before building any visuals, the exploration was framed around five guiding questions:

1. Which clubs gained or lost the most from VAR overall?

2. What kinds of incidents does VAR review most often?

3. Does the venue (home vs away) influence whether a review favours a team?

4. When during a match do VAR incidents tend to happen?

5. Do the purely judgment-based ("subjective") decisions tell the same story as the full set of decisions, or a different one?

Answering these required cleaning both tables in Power Query first, since the raw Date and Incident columns were not analysis-ready.

2.1 Cleaning the Date column

The raw Date values (e.g. "Aug. 17") had no year, and a few month abbreviations ("Sept.") broke Power Query's default date conversion. Since the season runs across a calendar-year boundary, the year was inferred from the month: August–December was treated as 2019 and January–July as 2020. Two rows recorded only "July" with no day and were left blank rather than guessed, since a day cannot be safely assumed.

```
let
    d = Text.Trim([Date]),
    month = Text.BeforeDelimiter(d, " "),
    day = try Number.FromText(Text.AfterDelimiter(d, " ")) otherwise null,
    year = if List.Contains({"Aug.","Sep.","Oct.","Nov.","Dec."}, month)
           then 2019
           else 2020
in
    if day = null then null
    else #date(year, Date.Month(Date.FromText("1 " & month & " 2000")), day)
```

The VAR used column was also checked at this stage: its two values, FOR and AGAINST, are text labels rather than a Boolean condition, so it was kept as Text throughout rather than converted to True/False.

	Column1	Column2	Column3	Column4	Column5	Column6	Column7
1	Team	Opponent Team	Date	Site	Incident	Time	VAR used
2	Brighton & Hove Albion	West Ham	Aug. 17	H	Brighton goal for Leandro Trossard ruled out for offside by Dan Burn, ...	27th minute	AGAINST
3	Brighton & Hove Albion	Aston Villa	Oct. 19	A	Conor Hourihane goal ruled out for a foul by Wesley on Mat Ryan	42nd minute	FOR
4	Brighton & Hove Albion	Everton	Oct. 26	H	Penalty awarded (scored by Neil Maupay) for foul on Aaron Connolly b...	78th minute	FOR
5	Brighton & Hove Albion	Leicester	Nov. 23	H	Jamie Vardy penalty retaken after James Maddison scored the rebound...	80th minute	FOR
6	Brighton & Hove Albion	Arsenal	Dec. 5	A	David Luiz goal disallowed for offside	63rd minute	FOR
7	Brighton & Hove Albion	Sheffield United	Dec. 21	H	John Egan goal ruled out for handball	8th minute	FOR
8	Brighton & Hove Albion	Sheffield United	Dec. 21	H	Jack O'Connell goal disallowed for offside	54th minute	FOR

Fig. 1 — VAR_Incidents_Stats before cleaning

	Team	Opponent Team	Date	Site	Incident type	Minute	VAR used
1	Brighton & Hove Albion	West Ham	8/17/2019	H	Offside	27	AGAINST
2	Brighton & Hove Albion	Aston Villa	10/19/2019	A	Foul	42	FOR
3	Brighton & Hove Albion	Everton	10/26/2019	H	Penalty	78	FOR
4	Brighton & Hove Albion	Leicester	11/23/2019	H	Penalty	80	FOR
5	Brighton & Hove Albion	Arsenal	12/5/2019	A	Offside	63	FOR
6	Brighton & Hove Albion	Sheffield United	12/21/2019	H	Handball	8	FOR
7	Brighton & Hove Albion	Sheffield United	12/21/2019	H	Offside	54	FOR
8	Brighton & Hove Albion	Tottenham	12/26/2019	A	Offside	24	FOR

Fig. 2 — VAR_Incidents_Stats after cleaning: Date resolved, Minute isolated as a number

2.2 Building the Incident type column

The original Incident column is valuable but too varied as free text to use directly as a chart category, so a new Incident type column was derived from it with a chain of Text.Contains checks. Reviewing every one of the 218 descriptions surfaced two issues with an earlier draft of this logic, both corrected in the final version below:

- An “Encroachment” branch was dropped: all 8 descriptions containing “encroachment” also contain “penalty” (encroachment only occurs on a penalty retake), so with Penalty checked first that branch could never be reached. Those incidents are classified as Penalty, which also better reflects the substance of the review.
- The Red Card check was moved ahead of Penalty: two incidents combine a dismissal with a penalty (e.g. “...sent off for deliberate handball on the goalline, penalty scored by Willian”). Checking Red Card first classifies these by their more severe outcome rather than letting the penalty branch catch them first.

```
if Text.Contains([Incident], "red card", Comparer.OrdinalIgnoreCase)
     or Text.Contains([Incident], "sent off", Comparer.OrdinalIgnoreCase) then "Red Card"
else if Text.Contains([Incident], "penalty", Comparer.OrdinalIgnoreCase) then "Penalty"
else if Text.Contains([Incident], "handball", Comparer.OrdinalIgnoreCase) then "Handball"
else if Text.Contains([Incident], "offside", Comparer.OrdinalIgnoreCase) then "Offside"
else if Text.Contains([Incident], "push", Comparer.OrdinalIgnoreCase)
     or Text.Contains([Incident], "foul", Comparer.OrdinalIgnoreCase) then "Foul"
else "Other"
```

2.3 Data model

VAR_Team_Stats and VAR_Incidents_Stats were linked on Team as a one-to-many relationship, with cross-filter direction set to Both. This was a deliberate choice: with only two tables in the model there is no risk of ambiguous filter paths, and Both is required for slicers such as Incident type and Site — which live only in the incident-level table — to also filter the club-level totals shown in the KPI cards and Net goal score chart.

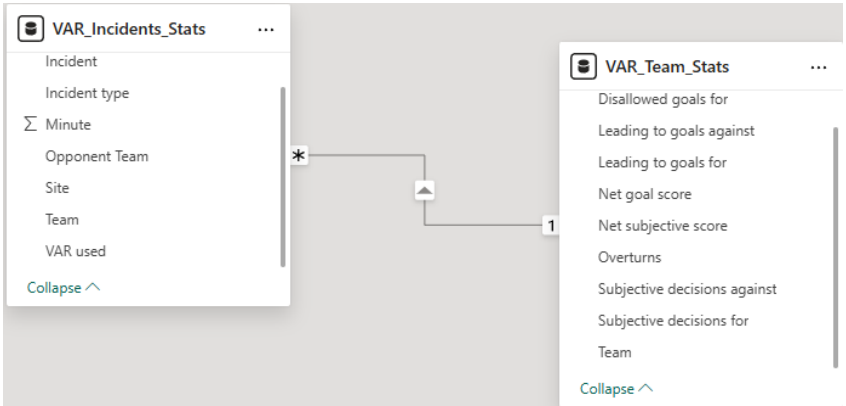


*Fig. 3 — Model view: * -to-1 relationship between VAR_Incidents_Stats and VAR_Team_Stats*

2.5 Enhancing charts with detailed tooltips

To let a reader see the underlying incidents behind any bar without leaving the page, a DAX measure was added to concatenate every matching incident description into a single tooltip string, separated by blank lines for readability:

-

```
All Incidents = CONCATENATEX(
    VALUES(VAR_Incidents_Stats[Incident]),
    VAR_Incidents_Stats[Incident],
    UNICHAR(10) & UNICHAR(160) & UNICHAR(10)
)
```

UNICHAR(160) inserts a non-breaking space between the line breaks — without it, Power BI collapses consecutive blank lines in tooltips into one. This measure was applied to the Tooltips field of four charts: Net Goal Score by Team, Total VAR Overturns by Team, Total VAR Decisions by Team: For vs Against, and Breakdown of VAR Goal Impact by Team, so a reader can hover over any club and read every incident that contributed to its numbers.

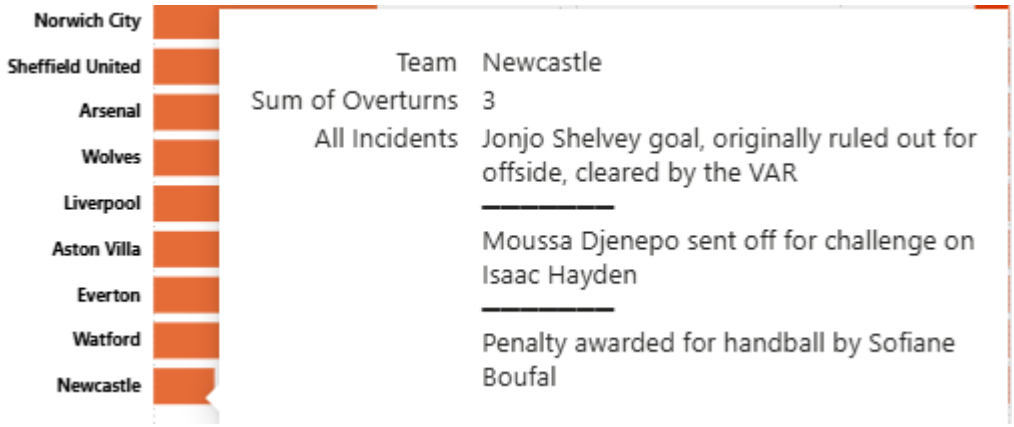
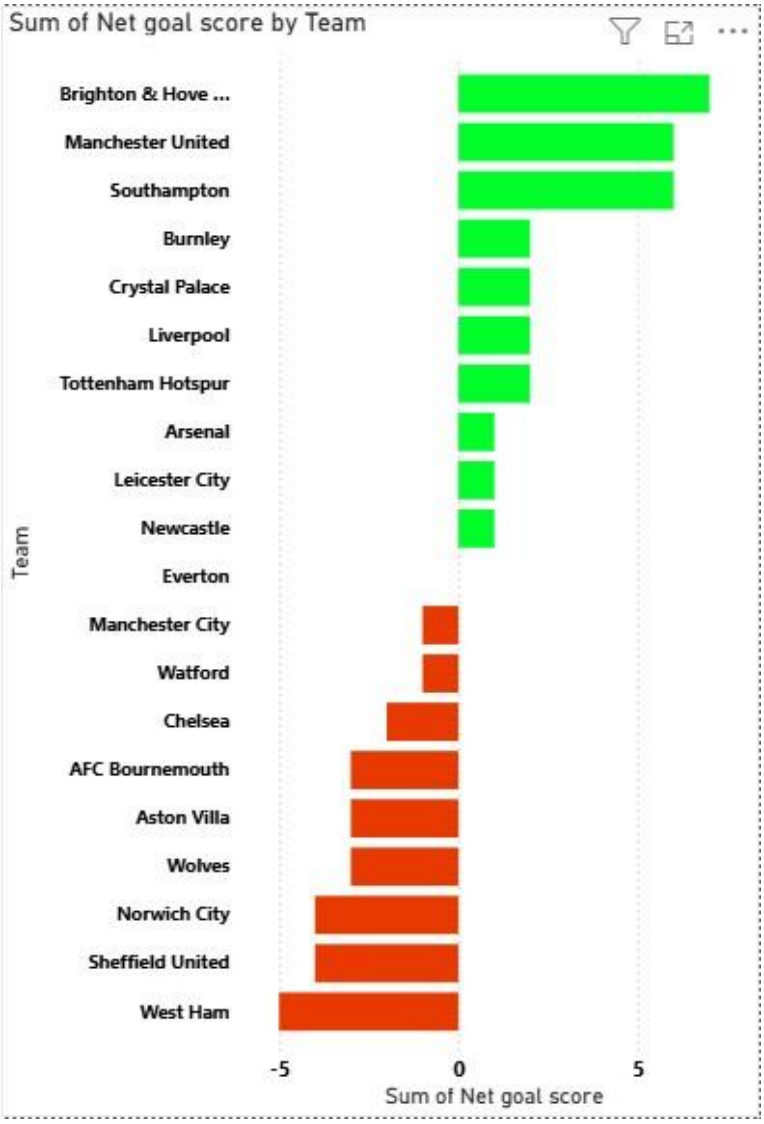


Fig. 7 — Tooltip showing concatenated incident details on hover

3. Visualizations

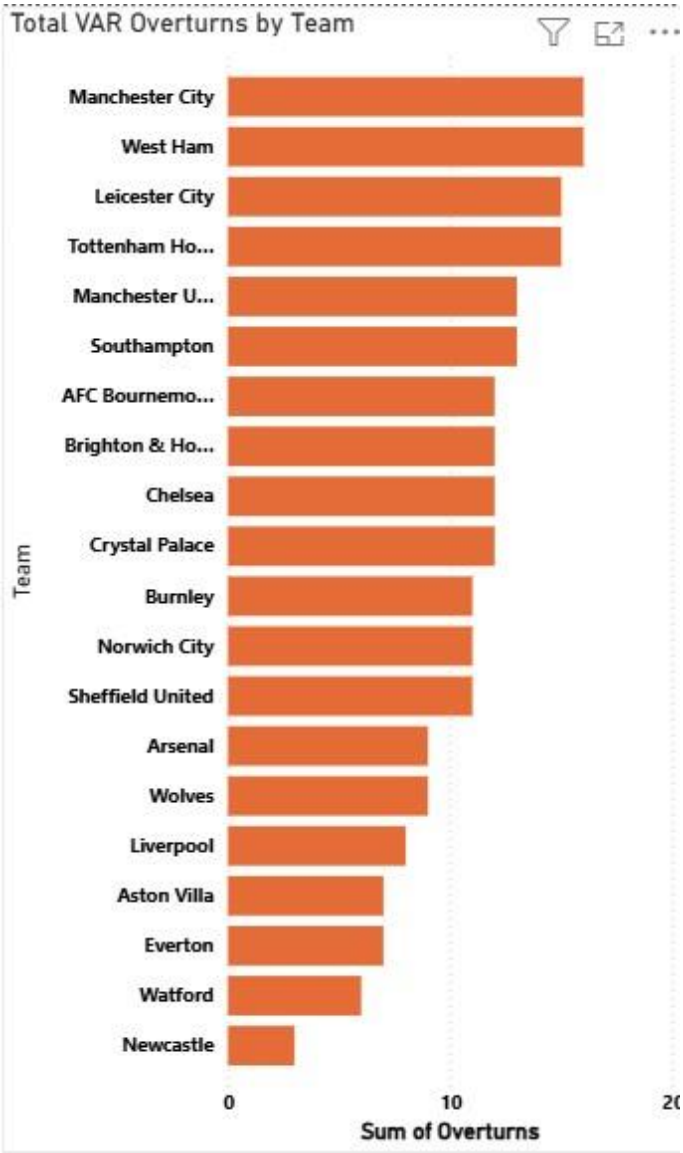
Nine visuals were built across four dashboard pages to answer the five guiding questions. Each is shown below with what it reveals.

3.1 Net Goal Score by Team



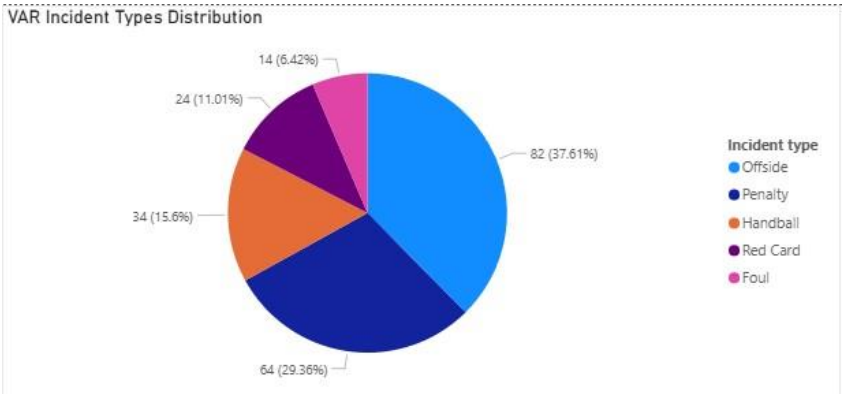
This is the headline club-level view: how many net goals each club gained or lost across the season because of VAR. Brighton & Hove Albion (+7), Manchester United (+6) and Southampton (+6) sit at the favourable end; West Ham United (−5), Sheffield United (−4) and Norwich City (−4) sit at the unfavourable end. Ten clubs finished with a positive net score, Everton was exactly neutral, and nine finished negative.

3.2 Total VAR Overturns by Team



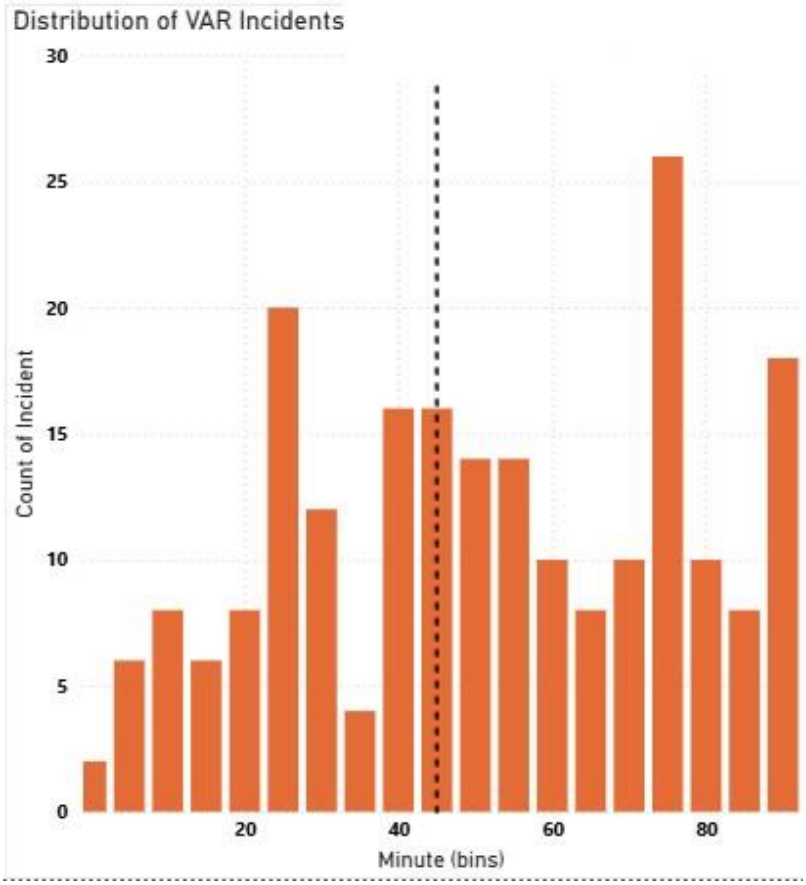
This counts how often each club was involved in a VAR review at all, regardless of outcome. Manchester City and West Ham were involved most often (16 each), while Newcastle was involved least (3). Read together with 3.1, this shows that being reviewed often is not the same as benefiting: Manchester City's 16 reviews still netted only −1.

3.3 VAR Incident Types Distribution



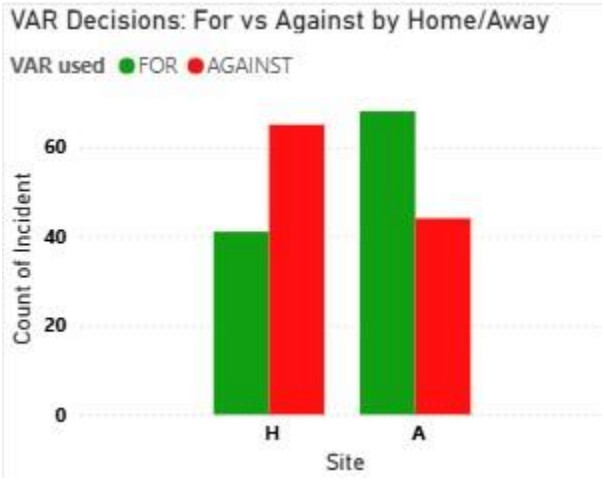
Across all 218 incidents, Offside is the single largest category (82, 37.6%), followed by Penalty (64, 29.4%), Handball (34, 15.6%), Red Card (24, 11.0%) and Foul (14, 6.4%). Offside and Penalty together account for two-thirds of all VAR activity, meaning the majority of reviews concern relatively fact-checkable calls rather than the more contentious foul/handball judgment calls.

3.4 Distribution of VAR Incidents by Match Minute



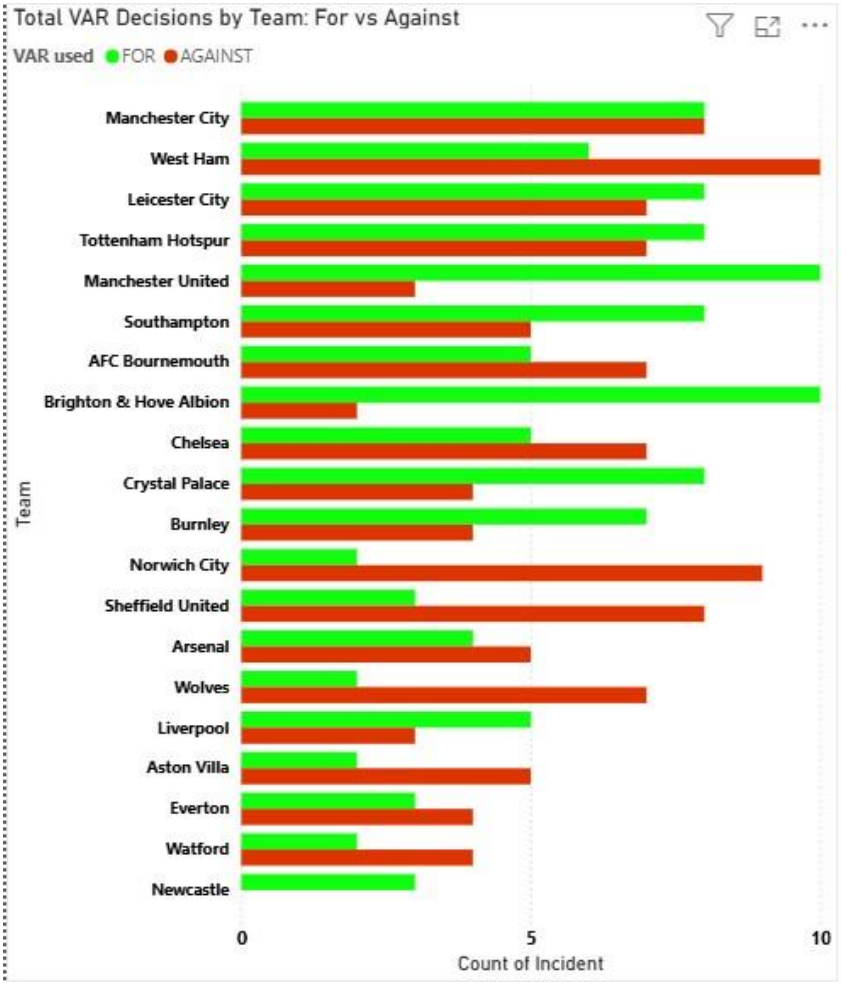
Incidents are binned into 5-minute windows across the 90+ minutes of play. The busiest window is minutes 75–80 (26 incidents), with secondary peaks around minutes 20–25 and 85–90. Incidents are not evenly spread — they concentrate in the closing stages of each half, when fatigue and rising stakes tend to produce more marginal, review-worthy contact.

3.5 VAR Decisions: For vs Against by Home/Away



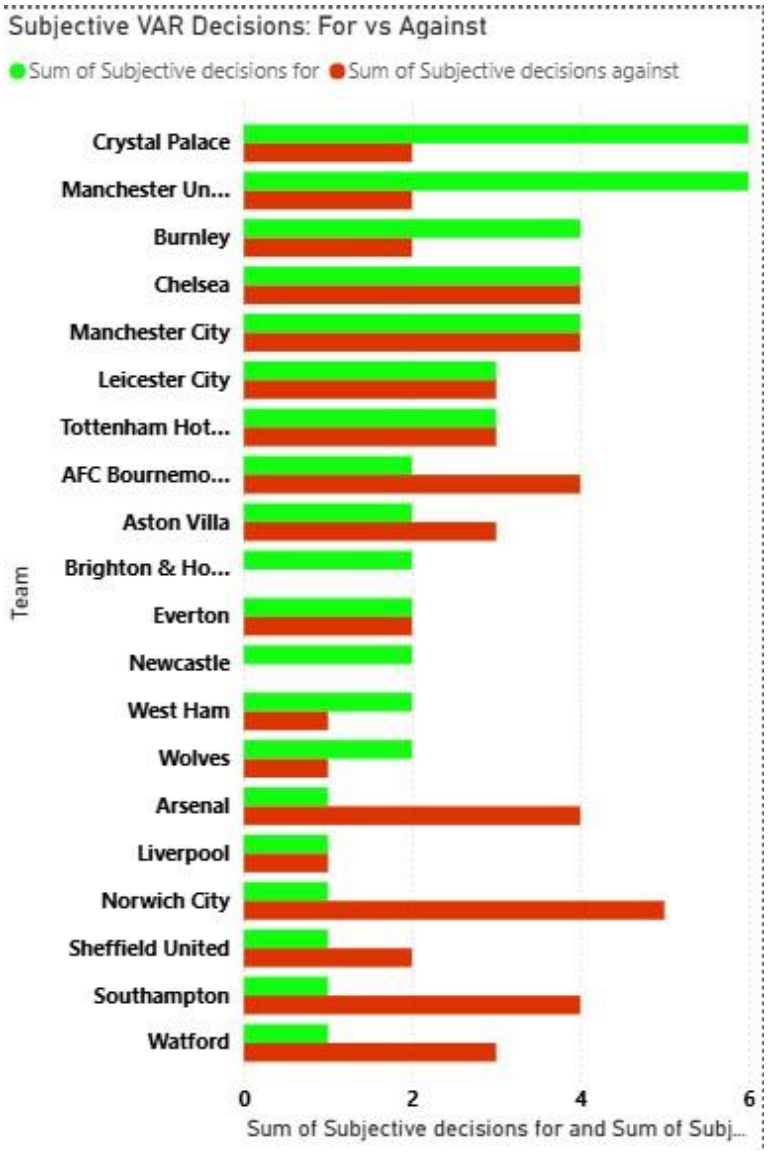
Splitting every decision by venue produces a counter-intuitive result: home teams received more unfavourable rulings than favourable ones (41 for vs 65 against), while away teams received more favourable rulings than unfavourable ones (68 for vs 44 against). Traditional home advantage does not appear to extend to VAR outcomes in this dataset — if anything, the pattern runs the other way.

3.6 Total VAR Decisions by Team: For vs Against



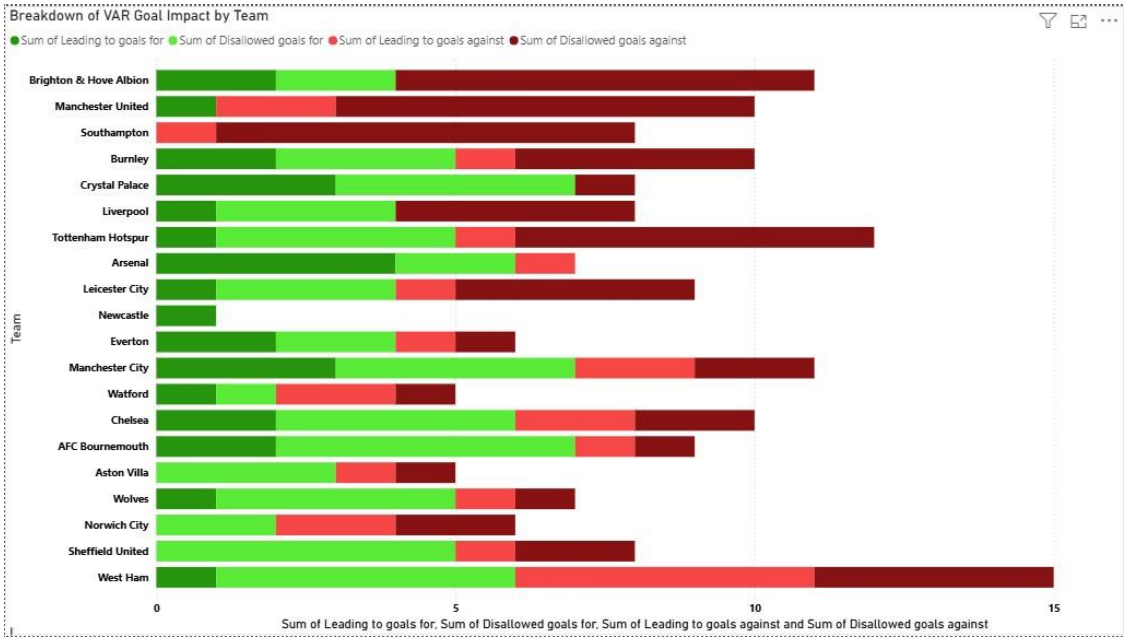
This breaks every decision (not just the net score) into its FOR and AGAINST components per club. Manchester United and Brighton stand out for how lopsided their split is toward FOR; West Ham, Norwich City and Sheffield United show the opposite pattern, with AGAINST clearly dominating.

3.7 Subjective VAR Decisions: For vs Against



Narrowing to only the judgment-based decisions (fouls, handballs, red cards — excluding offside) tells a partly different story. Crystal Palace and Manchester United have the best subjective split (6 for vs 2 against each), while Norwich City has the worst (1 for vs 5 against). Some clubs that looked balanced in the full total, such as Arsenal, show a much more one-sided picture once only the subjective calls are considered.

3.8 Breakdown of VAR Goal Impact by Team



This decomposes each club's Net goal score (3.1) into its four building blocks — goals led to or disallowed, for and against. It shows why two clubs can reach a similar net score through very different routes: Arsenal's modest +1 comes almost entirely from goals led to for the club, with very little disallowed against it, whereas Tottenham Hotspur's more modest total masks a much larger volume of decisions in both directions.

3.9 KPI Summary Cards



Three headline numbers anchor the Overview page: 20 clubs, 218 total incidents, and 218 total overturns (the two figures match because every logged incident is, by definition, a reviewed/overturned decision).

4. Key Findings

1. Brighton & Hove Albion gained the most from VAR (net +7) and West Ham United lost the most (net −5) — a 12-goal swing between the two extremes of the table.

2. Two-thirds of all VAR activity is Offside (37.6%) or Penalty (29.4%); the more judgment-heavy categories — Handball, Red Card and Foul — combine for just under a third of incidents.

3. VAR outcomes leaned toward away teams league-wide: away sides received a favourable ruling 61% of the time (68 for vs 44 against), while home sides received one only 39% of the time (41 for vs 65 against).

4. Incidents cluster late in each half. The single busiest window was minutes 75–80 (26 incidents), suggesting review-worthy contact rises as fatigue and match stakes increase.

5. Being reviewed often is not the same as benefiting. Manchester City and West Ham were involved in the most reviews of any club (16 each), yet finished at −1 and −5 respectively — high VAR involvement did not translate into a favourable net outcome for either club.

6. The subjective (judgment-call) breakdown does not always mirror the full total. Crystal Palace and Manchester United had the best subjective net split, and Norwich City the worst, showing that part of the season's VAR narrative sits specifically in foul/handball/red-card calls rather than in fact-based rulings like offside.

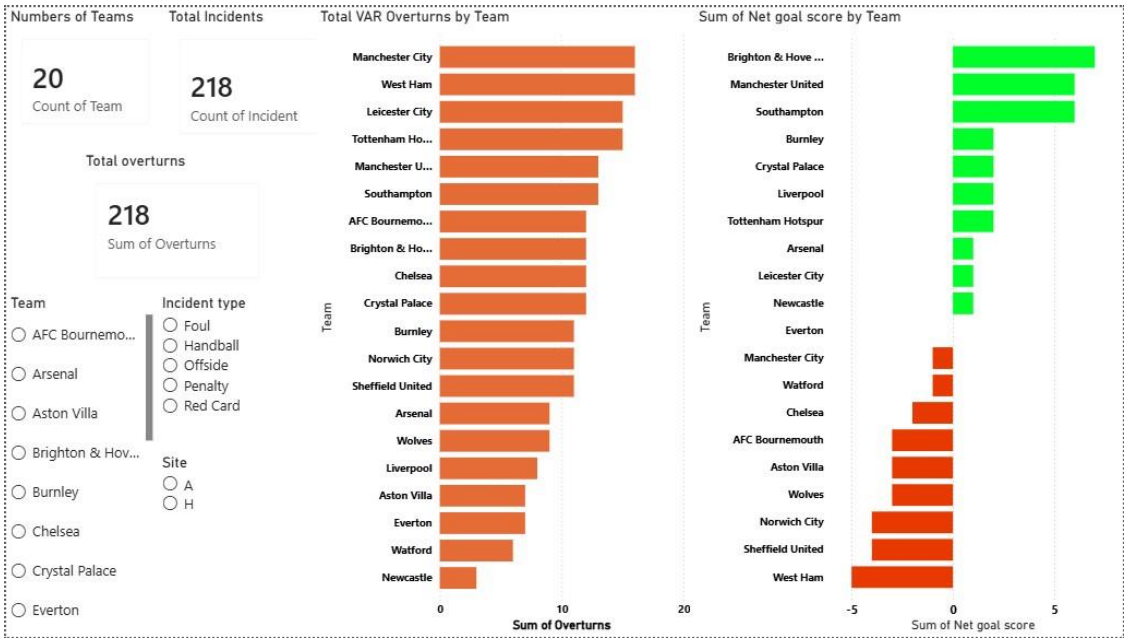
5. Dashboard Narrative

The findings above are delivered as a four-page, fully interactive Power BI dashboard rather than a set of static charts. The pages are designed to be read in order, moving from a broad summary to increasingly specific detail:



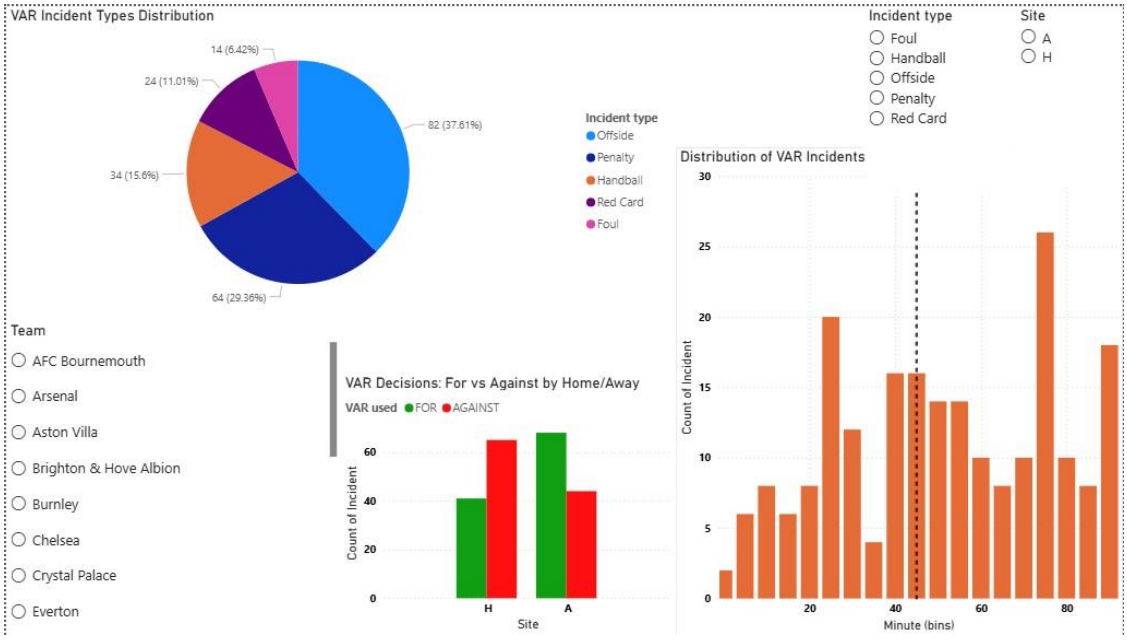
5.1 Overview

The entry page. Three KPI cards (Teams, Total Incidents, Total Overturns) give the headline scale of the dataset at a glance, sitting beside the Net Goal Score and Total Overturns charts so the two central “who benefited / who was involved” questions are answered immediately. Team, Incident type and Site slicers sit on the left, ready to filter everything that follows.



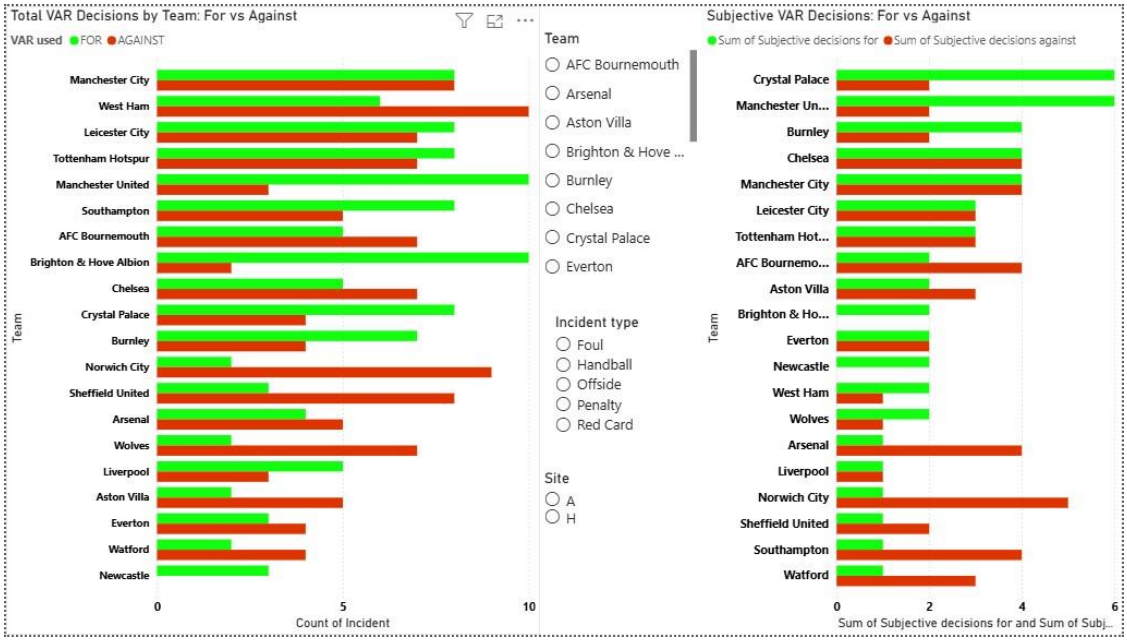
5.2 Details

Moves from “who” to “what and when”: the Incident Types pie, the Match Minute distribution, and the Home/Away split sit together so a reader can see what kind of incidents dominate and where in the match — and at which venue — they concentrate.



5.3 Total & Subjective VAR Decisions

Places the full For/Against split next to the narrower subjective-only split for every club, side by side, so the reader can directly compare a club's overall VAR record against the part of that record driven purely by human judgement.



5.4 Goal Impact Breakdown

The final, most granular page: the four components behind every club's Net goal score, stacked so the reader can see not just the total but where it came from.



5.5 Interactivity

All three slicers (Team, Incident type, Site) are synced across all four pages using Power BI's Sync Slicers pane, and are kept visible on the Overview page only. Selecting a single club or incident type on the Overview page therefore filters every chart on every subsequent page, letting a reader follow one club's or one incident type's story end-to-end — from its headline net score down to the individual goal-impact components — without leaving the dashboard.